



InsightMart

Frontend Documentation

Architecture, data flow, application pages
and demo login credentials

Product	InsightMart - Sales Analytics & Reporting Platform
Scope	React single-page frontend (10 pages, 71 source files)
Stack	React 18 - Vite 5 - Tailwind CSS 3 - Recharts - Framer Motion
Backend	Flask REST API + MySQL (contract implemented, mock-backed)
Data mode	Mock backend enabled via VITE_USE MOCK

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Read this first

The Flask backend is not required to run or review this frontend. Every page is served by an in-browser mock backend that mirrors the documented REST contract, so the entire application is navigable immediately. Section 8 lists the three demo accounts that unlock each role.

SECTION 1

Technology Stack

The frontend is a decoupled React single-page application. It owns routing, state and presentation, and talks to Flask over JSON with a JWT.

1.1 Core libraries (from the project specification)

LIBRARY	VERSION	PURPOSE IN THIS BUILD
React	18.3	Component model for all pages, layouts and primitives
React Router DOM	6.28	Client-side routing, nested layouts and route guards
Axios	1.7	HTTP client with a JWT request interceptor and 401 handling
Recharts	2.15	Bar, line, histogram and donut charts on Dashboard and Analysis
Context API	built-in	Auth session, theme and toast notifications
React Hook Form	7.54	Validation on login, register, invite and account forms
Tailwind CSS	3.4	Utility styling driven by a custom design-token layer

1.2 Added for build quality

Three additions sit outside the original library list. They were added deliberately, to meet the requirement for a modern, animated interface without hand-rolling the same behaviour repeatedly.

LIBRARY	VERSION	WHY IT WAS ADDED
Framer Motion	11.15	Route transitions, scroll reveals, shared-layout indicators, count-up figures
Lucide React	0.468	Consistent line icon set - the interface contains no emoji
Papa Parse	5.4	Client-side CSV preview so columns are shown before upload
Vite	5.4	Dev server and production bundler
clsx + tailwind-merge	2.x	Conditional class composition without specificity conflicts

1.3 Backend contract this frontend targets

The service layer is written against the documented Flask stack: Flask-JWT-Extended for tokens, Flask-CORS for cross-origin access, Pandas for CSV parsing and analytics, Marshmallow for validation, ReportLab for PDF export and Flask-SQLAlchemy over MySQL 8. No frontend change is required when that backend comes online.

SECTION 2

System Architecture

Three classes of user reach the same React application. It calls one Flask REST API, which owns all business logic and the single MySQL database.

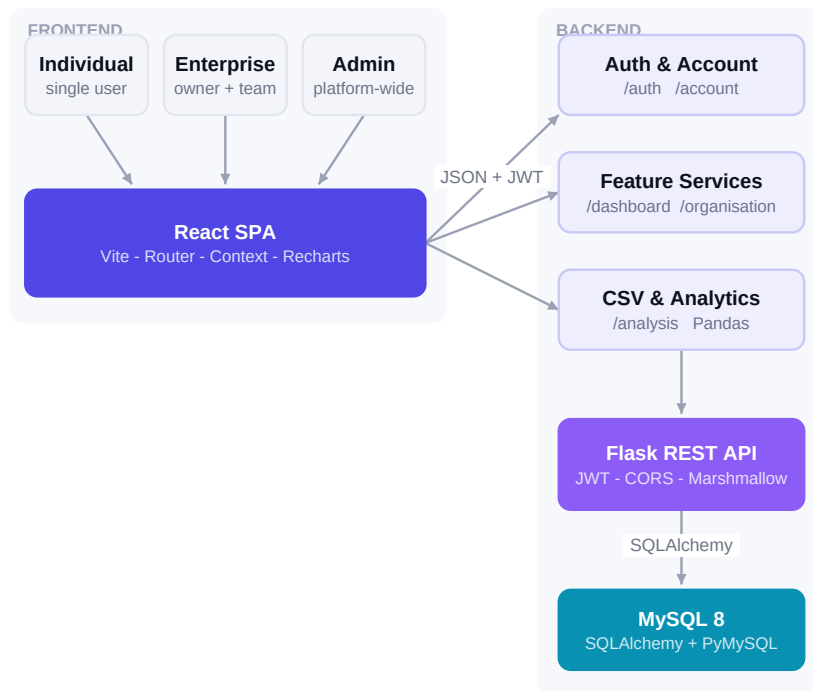


Figure 1 - System architecture. The frontend never reaches the database directly.

2.1 Boundaries

- **Presentation** lives entirely in React. No business rule is duplicated on the client beyond role gating, which exists for navigation only and is re-enforced by the API.
- **Authentication** is a JWT issued by Flask, held in the auth context and attached to every request by an Axios interceptor. A 401 response clears the session automatically.
- **Analytics** are computed server-side by Pandas. The frontend receives finished figures and is responsible only for rendering them.
- **The database** is reachable only through SQLAlchemy inside Flask.

SECTION 3

Frontend Architecture

Inside the SPA, concerns are separated into a provider tree, a routing layer, layout shells, pages and a single service gateway.

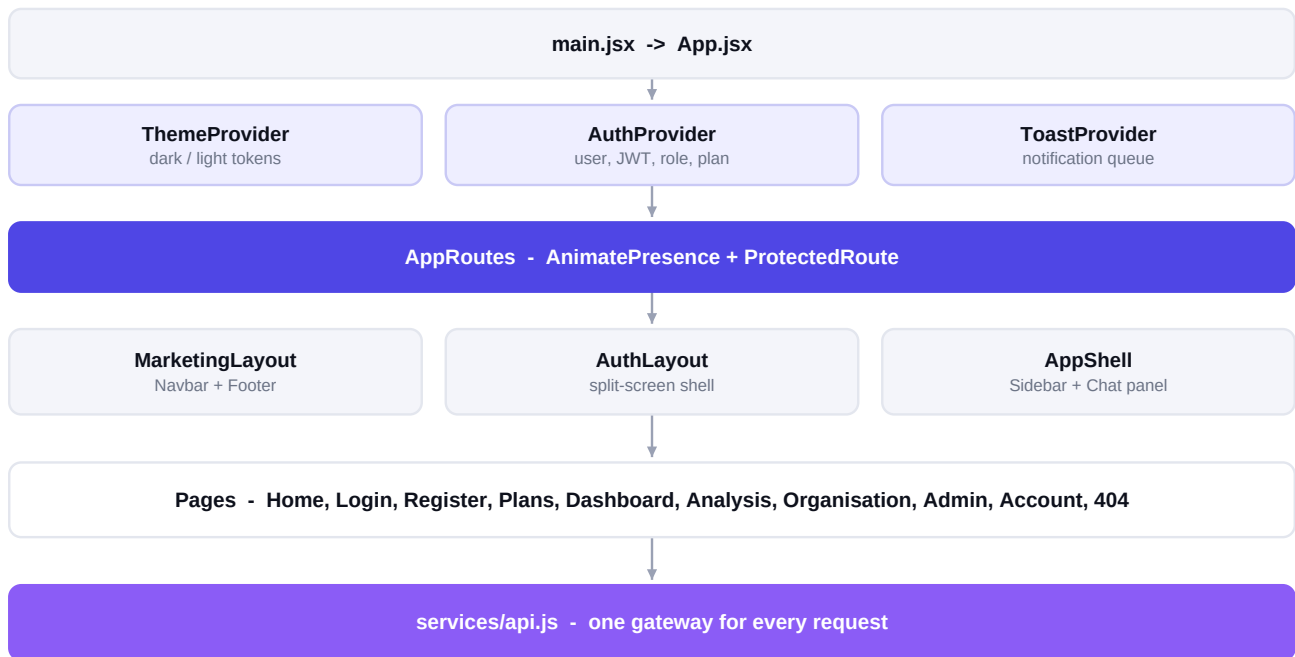


Figure 2 - Frontend composition, outermost layer at the top.

3.1 The service gateway

Components never import Axios. They call named functions on `services/api.js`, which decides at call time whether to hit Flask or the in-browser mock. This is the single switch that makes the whole application reviewable without a backend.

GROUP	FUNCTIONS	FLASK ENDPOINT
authApi	login, register, forgotPassword, logout	/auth/*
accountApi	getProfile, updateProfile, changePassword, changePlan	/account/*
dashboardApi	get	/dashboard
analysisApi	get, listUploads, upload, exportReport	/analysis/*
organisationApi	get, invite, updateRole, remove	/organisation/*
adminApi	overview, setUserStatus	/admin/*

GROUP	FUNCTIONS	FLASK ENDPOINT
chatApi	send	/chat/message

Switching to the real backend

Set VITE_USE MOCK=false and point VITE_API_BASE_URL at the Flask host. No component changes.

SECTION 4

Project Structure

The source tree as built - 71 files under src/.

```
insightmart-frontend/
|-- index.html           fonts, theme boot script (no flash of wrong palette)
|-- vite.config.js       @/ alias onto src/
|-- tailwind.config.js   token-backed colours, fluid type scale, keyframes
|-- .env                 VITE_API_BASE_URL, VITE_USE MOCK
`-- src/
    |-- main.jsx App.jsx entry and provider tree
    |-- styles/index.css design tokens, glass/rim primitives, utilities
    |-- lib/
    |   |-- cn.js        class merge helper
    |   |-- motion.js    easings, springs, shared variants
    |   |-- formatters.js currency, compact numbers, percent, dates
    |   |-- constants.js roles, nav, plans, plan matrix
    |-- hooks/           useCountUp, useTilt, useMediaQuery, useLocalStorage,
    |                   useOnClickOutside, useLockBodyScroll
    |-- context/
    |   |-- AuthContext.jsx user, token, role, plan, permissions
    |   |-- ThemeContext.jsx dark / light / system
    |   |-- ToastContext.jsx animated notification queue
    |-- services/
    |   |-- api.js        the single gateway (Axios + mock switch)
    |   |-- mock/
    |       |-- mockData.js seeded dataset - 260 sales records and more
    |       |-- mockApi.js  latency-simulated resolver per endpoint
    |-- routes/AppRoutes.jsx route table with animated transitions
    |-- components/
    |   |-- Navbar Sidebar ChatSidebar (specified components)
    |   |-- KpiCard ChartCard ProtectedRoute (specified components)
    |   |-- home/ Hero, HeroMockup, FeatureBento, HowItWorks,
    |   |   StatsBand, LogoMarquee
    |   |-- layout/ MarketingLayout, AuthLayout, AppShell, Footer, ScrollToTop
    |   |-- motion/ Reveal, Stagger, Spotlight, MagneticButton,
    |   |   AuroraBackground, GridBackdrop, PageTransition, CountUp
    |   |-- charts/ RevenueLineChart, SalesBarChart, CategoryDonut,
    |   |   OrderHistogram, RetentionChart, Sparkline, chartTheme
    |   |-- ui/ Button, Card, Badge, Input, Select, Tabs, Table, Modal,
    |   |   Skeleton, Progress, EmptyState, FileDropzone, ThemeToggle, Logo
    |-- pages/
    |   Home, Login, Register, MyAccount, Dashboard, CsvAnalysis,
    |   Organisation, Admin, Plans, NotFound
```

SECTION 5

Data Flow

The journey every account follows, from creating an account through to exporting a report.

STEP	STAGE	WHAT HAPPENS
1	Sign up / log in	Register as Individual or Enterprise. Flask hashes the password and returns a JWT, which the auth context stores.
2	Organisation setup	Enterprise accounts create an organisation and invite team members. Individual accounts skip this step entirely.
3	CSV upload	The user drops a sales file on the CSV Analysis page. It is previewed client-side before anything is sent.
4	Validation & parsing	Flask checks the file type and required columns, then reads it with Pandas.
5	Cleaning & storage	Missing values are handled, records are normalised and written to MySQL through SQLAlchemy.
6	Analytics computation	The analytics service computes revenue, top products and regions, profitability, retention and order metrics.
7	Visualisation	Dashboard and CSV Analysis render the results as KPI cards, bar, line, histogram and donut charts.
8	Export & support	The user exports a ReportLab PDF or asks the chat sidebar for help. Admin oversees all of it at any point.

5.1 Lifecycle of a single request

Every call in the application follows the same path, whichever backend is active.

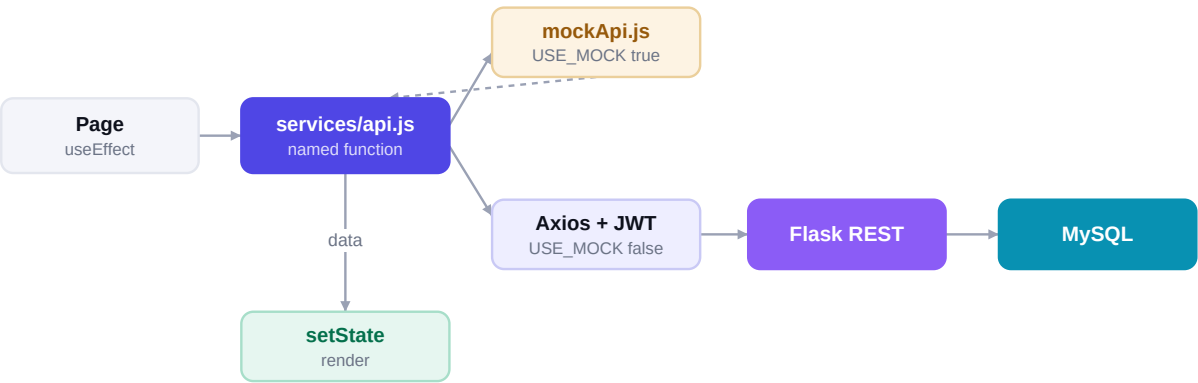


Figure 3 - One gateway, two possible backends, identical response shape.

Why the mock matters

The mock resolvers return exactly the shape the Flask endpoints are specified to return, and add a short artificial latency. That keeps loading skeletons, error states and empty states honest rather than decorative - they are exercised on every page load.

SECTION 6

Application Pages

Ten routes: four public, six behind authentication. Each is listed with its access rule and what it actually contains.

PAGE	ROUTE	ACCESS	CONTENTS
Home	/	Public	Full-viewport hero with animated headline and a live product mockup; technology marquee; eight-tile feature grid; four-step 'how it works' rail with a scroll-driven progress line; statistics band; plan preview; closing call to action.
Login	/login	Public	Split-screen form with validation and password reveal, an animated error banner, and a demo account picker that fills and submits the form in one click (see Section 8).
Register	/register	Public	Individual / Enterprise segmented control with a sliding indicator. Choosing Enterprise reveals a required organisation field. Live password-strength meter, confirmation matching, terms acceptance.
Plans	/plans	Public	Monthly / annual toggle with morphing prices, three tier cards, a nine-row feature comparison matrix and a five-item FAQ accordion.
Dashboard	/dashboard	All signed-in	Four KPI cards with count-up values and sparklines; revenue and orders trend; category donut; regional revenue against profit; top products table; recent uploads; gross profit, margin and repeat-rate strip. Date-range filter re-queries and re-animates the charts.
CSV Analysis	/analysis	All signed-in	Drag-and-drop upload with progress ring and client-side column preview, then a staged processing sequence. Results: five KPIs, sales by month/quarter/year, revenue and order trend, order-value histogram, retention, category share, top products and branch profitability. PDF export.
Organisation	/organisation	Enterprise, Team, Admin	Organisation header, active/pending/upload summary tiles, seat usage meter, member table with inline role editing and removal, invite modal, and a pending invitations list.
Admin	/admin	Admin only	Four platform statistics; searchable and role-filterable user table with suspend and reinstate; organisations table; subscription distribution donut; and a platform-wide upload feed including failures and their reasons.
My Account	/account	All signed-in	Profile, Security and Subscription tabs with an animated indicator. Profile details, identity card, password change, usage meters and an in-place plan switcher.
Not Found	*	Public	Role-aware 404 with return links to home, and to the dashboard when signed in.

6.1 Chart coverage

CHART TYPE	USED FOR	APPEARS ON
Bar	Sales by month, quarter and year; top products; categories; regions	Dashboard, CSV Analysis
Line / area	Revenue and order trends over time	Dashboard, CSV Analysis, Hero
Histogram	Distribution of order values	CSV Analysis
Donut	Share of sales by category or region; subscription split	Dashboard, CSV Analysis, Admin
Stacked area	New against returning customers	CSV Analysis
Sparkline	Inline trend inside KPI cards	Dashboard

SECTION 7

User Roles and Access

Two account types plus a team role and a platform-wide administrator.

ROLE	DESCRIPTION
Individual	A single-person account. Reaches Home, Authentication, My Account, Dashboard and CSV Analysis. No organisation or team features.
Enterprise	Creates an organisation, invites and manages team members, and shares dashboards and analysis through the Organisation page.
Team member	Invited into an Enterprise organisation by its owner. Access is scoped to that organisation's data.
Admin	Platform-level role with full access. Manages all users, organisations, uploads and subscription plans through the Admin page.

7.1 Route permission matrix

ROUTE	INDIVIDUAL	ENTERPRIS E	TEAM	ADMIN
/ , /plans , /login , /register	Yes	Yes	Yes	Yes
/dashboard	Yes	Yes	Yes	Yes
/analysis	Yes	Yes	Yes	Yes
/account	Yes	Yes	Yes	Yes
/organisation	-	Yes	Yes	Yes
/admin	-	-	-	Yes

Gating is implemented by `ProtectedRoute`. An unauthenticated visitor is sent to `/login` with the attempted path remembered, so signing in returns them where they were headed. A signed-in user without the required role is sent to `/dashboard` instead - they are authenticated, simply not permitted there. The sidebar renders only the entries the current role may reach.

SECTION 8

Demo Login Credentials

Three accounts are seeded into the mock backend, one per role, so every gated page can be reached without a database.

One password for all three accounts

demo1234

These are throwaway credentials for a local mock backend. They are not real secrets and grant no access to any live system.

ROLE	EMAIL	PASSWORD	SIGNED IN AS	UNLOCKS
Individual	individual@insightmart.dev	demo1234	Ananya Rao	Dashboard, CSV Analysis, My Account, Plans
Enterprise	enterprise@insightmart.dev	demo1234	Vikram Shah	The above, plus the Organisation page
Admin	admin@insightmart.dev	demo1234	Priya Menon	Everything, including the Admin page

8.1 How to sign in

- Open `/login`. The three accounts are listed on the page itself as clickable cards under the heading *Demo accounts*.
- Selecting a card fills both fields and submits immediately, landing on the dashboard for that role.
- The shared password is displayed on the same panel with a copy button, so it can also be typed manually.
- Registering a new account through `/register` also works. It is added to the in-memory session store and can sign in straight away, using whatever password was chosen.

8.2 Plan and role assignment

ACCOUNT	ROLE	PLAN	ORGANISATION	UPLOADS / MONTH
Ananya Rao	individual	Free	None	6 of 30
Vikram Shah	enterprise	Enterprise	Northwind Retail Group	42 of 300
Priya Menon	admin	Enterprise	None	12 of 999

Seeing the Free-tier experience

Sign in as the Individual account to see premium blocks locked. On CSV Analysis, the order-value histogram, retention chart and branch profitability table are blurred behind an upgrade prompt, and PDF export is unavailable - exactly the Free limits described in Section 9.

8.3 Newly registered accounts

A registration made through the form is assigned a role from the account-type control: Individual becomes role `individual` on the Free plan; Enterprise becomes role `enterprise` on the Enterprise plan with the organisation name captured at signup. Session registrations are held in memory only and are cleared on a full page reload.

SECTION 9

Feature Plans

Three tiers control account limits and feature access. Free and Pro are self-serve; Enterprise adds organisation and team capabilities.

FEATURE	FREE	PRO	ENTERPRISE
Accounts	4-5 accounts	Higher / individual	Organisation + team members
CSV uploads	1 per day	Increased upload limit	Increased / shared uploads
Dashboard & charts	Basic charts & KPIs	Advanced charts & KPIs	Advanced charts & KPIs
CSV Analysis page	Core metrics only	Full analysis suite	Full analysis suite
PDF report export	Not included	Included	Included
Organisation / team page	Not available	Not available	Included
Chat sidebar support	Basic	Priority	Priority
Trial period	-	Premium features	Premium features
Payment gateway	Not required	Not required	Not required

Plan definitions live in one place - `src/lib/constants.js` - and drive the Plans page, the Home page preview, the My Account subscription tab, the sidebar upgrade prompt and the premium locks on CSV Analysis. Changing a limit there changes it everywhere.

SECTION 10

Design and Animation System

The visual language is defined once and consumed everywhere, so the interface stays consistent and themes correctly.

10.1 Colour and theme

Colours are CSS custom properties holding raw RGB channels, declared on `:root` with a `[data-theme="dark"]` override. Tailwind consumes them through a wrapper so opacity modifiers still work. A boot script in `index.html` resolves the stored preference before first paint, so there is no flash of the wrong palette. Every page supports dark and light.

TOKEN GROUP	PURPOSE
canvas / surface / elevated / sunken	Background depth layers
ink / muted / faint	Text hierarchy
brand / violet / cyan	Accent ramp used on CTAs, chart strokes and focus rings
success / warn / danger	KPI deltas, statuses, validation and margin badges
hairline	All borders and dividers, applied at low alpha

10.2 Typography

Sora for display headings, Inter for interface text and JetBrains Mono for identifiers. Headings use a fluid `clamp()` scale. Every figure in a KPI card, chart axis or table cell uses tabular numerals so digits align in columns and do not jitter while counting up.

10.3 Motion

BEHAVIOUR	WHERE
Route transition - fade and rise	Every page change, via <code>AnimatePresence</code>
Scroll reveal and stagger	Marketing sections, cards, table rows
Shared-layout indicator	Sidebar active pill, tab underlines, segmented controls
Count-up figures	KPI cards, statistics band, summary tiles
Self-drawing SVG paths	Hero mockup and auth panel charts
Cursor parallax and spotlight	Hero product window, feature and KPI cards
Magnetic buttons	Primary calls to action
Staged progress sequence	CSV upload: validate, parse, clean, compute

Reduced motion is respected

Every animated component checks `prefers-reduced-motion`. When it is set, decorative motion is removed, parallax and drift stop, and count-up figures snap straight to their final values. The interface remains fully usable and no information is lost.

SECTION 11

Running the Project

Node 20 or newer is required.

11.1 Install and start

```
cd insightmart-frontend
npm install
npm run dev          # http://localhost:5173

npm run build        # production bundle into dist/
npm run preview      # serve the built bundle
```

The code block above renders in white on the dark panel; copy it as plain text.

11.2 Environment variables

VARIABLE	DEFAULT	MEANING
VITE_API_BASE_URL	http://localhost:5000/api	Base URL of the Flask REST API
VITE_USE MOCK	true	When true the in-browser mock backend serves every request

11.3 Connecting the Flask backend

- Start Flask with CORS enabled for the Vite origin.
- Set `VITE_USE MOCK=false` in `.env` and restart the dev server.
- Confirm the endpoints in Section 3.1 return the documented response shapes.
- No component or page needs to change - only the gateway resolves differently.

11.4 Known follow-ups

- The production bundle is roughly 1.05 MB (308 KB gzipped), dominated by Recharts. Route-level code splitting is the obvious next optimisation.
- Password recovery is stubbed on the client and awaits the Flask endpoint.
- Session registrations are held in memory and clear on reload; this resolves once Flask persists users.

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